Spatiotemporal Change Detection

Using Deep Learning on Multi-Temporal Satellite Imagery

Faridpur Engineering College

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Course Code: CSE-4114

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Exam Roll: 57

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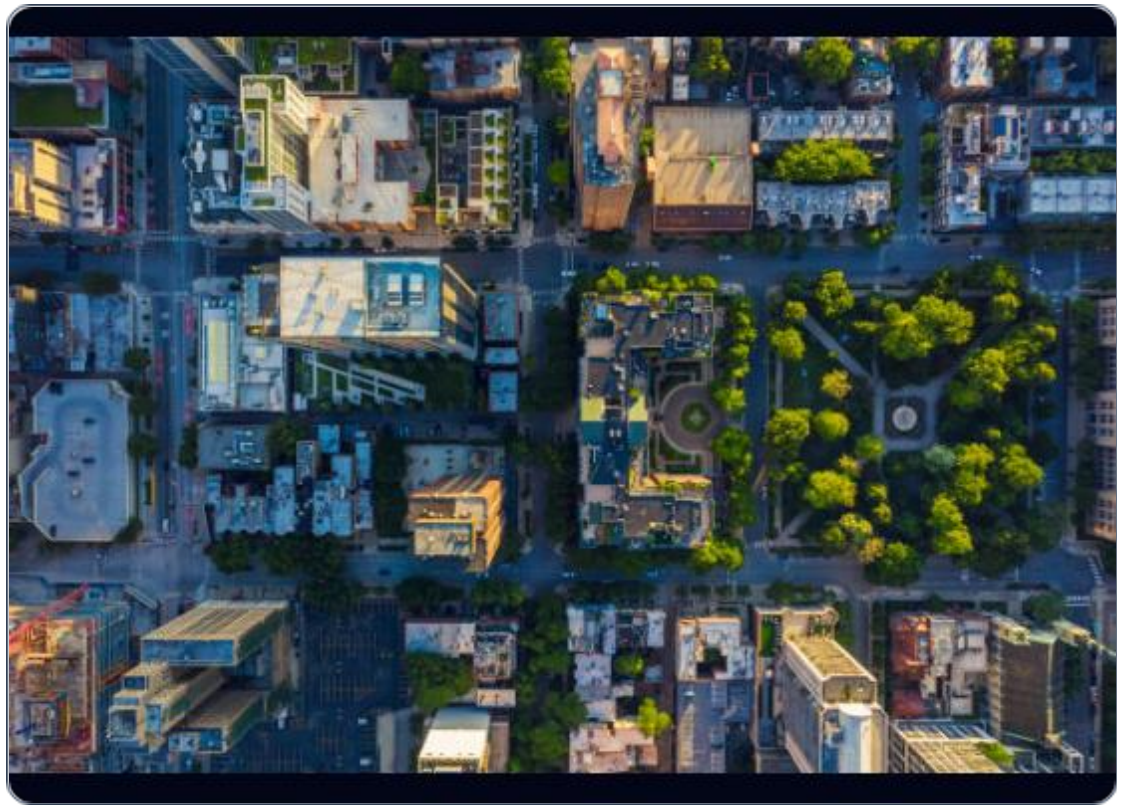
Introduction

- **Urban Monitoring:** Tracking infrastructure changes is essential for urban planning, disaster management.
- **Limits of Tradition:** Conventional manual mapping and pixel-differencing methods are time-consuming, expensive, and unreliable in complex environments.
- **Deep Learning Shift:** Convolutional Neural Networks (CNNs) have revolutionized automated feature extraction and pixel-level building segmentation.
- **Cost-Effective Approach:** This study focuses on creating robust solutions utilizing easily accessible RGB satellite imagery rather than expensive multispectral data.



Motivation

Rapid Urbanization

Continuous infrastructure development demands highly automated and scalable monitoring systems to track expansions efficiently across vast geographic regions.

Resource Constraints

Advanced multispectral satellite sensors are costly, leaving developing areas highly reliant on publicly accessible and more affordable RGB imagery.

Technical Challenges

Distinguishing true structural changes from environmental illusions (like shadows and seasonal vegetation shifts) requires highly advanced feature learning mechanisms.

Societal Impact

Automating this process helps authorities efficiently spot illegal constructions, assess disaster damages rapidly, and manage smart city growth comprehensively.

Problem Statement

Data Dependency Current high-performance detection models rely too heavily on rare, high-resolution multispectral datasets that are not universally accessible.

False Positives Variations in lighting, atmospheric noise, and seasonal changes frequently cause false detections and inaccuracies in traditional models.

Geographic Overfitting Models trained on specific geographical regions often fail to generalize to the architectural styles and varying densities of new, unseen cities.

The Gap There is a critical, unaddressed need for a cost-efficient spatiotemporal change detection system specifically optimized for standard RGB images.

Objective

Framework Development

Build a deep learning-based spatiotemporal framework for building change detection using multi-temporal imagery.

Benchmark Training

Utilize high-quality benchmark datasets to train the neural model comprehensively across diverse urban patterns.

Enhance Robustness

Improve the model's resistance to environmental noise, seasonal variations, and unexpected image distortions.

Practical Deployment

Validate the model using standard evaluation metrics on inexpensive RGB data to ensure high real-world viability.

Literature Review

Approach	Key Methods	Characteristics
Minaee et al. (2022)	Image Segmentation Using Deep Learning: A Survey	Conducted a large-scale review of deep learning segmentation models. Compared FCN, U-Net, DeepLab, and Transformer-based approaches.
Yu et al. (2022)	AGs-UNet: Attention Gates U-Net for Building Extraction	Modified U-Net by adding attention gates. The attention mechanism helps the model focus on important building regions.
Liu et al. (2024)	STransU2Net: Transformer-Based Hybrid Model	Proposed a hybrid CNN + Transformer segmentation model. CNN extracts local spatial features. Achieved better performance in complex urban scenes.
Liu et al. (2025)	Building Group Recognition Integrating Spatial and Semantic Similarity	Proposes a method to recognize building groups instead of individual buildings. Uses spatial relationships (how buildings are arranged) and semantic similarity (feature similarity between buildings).

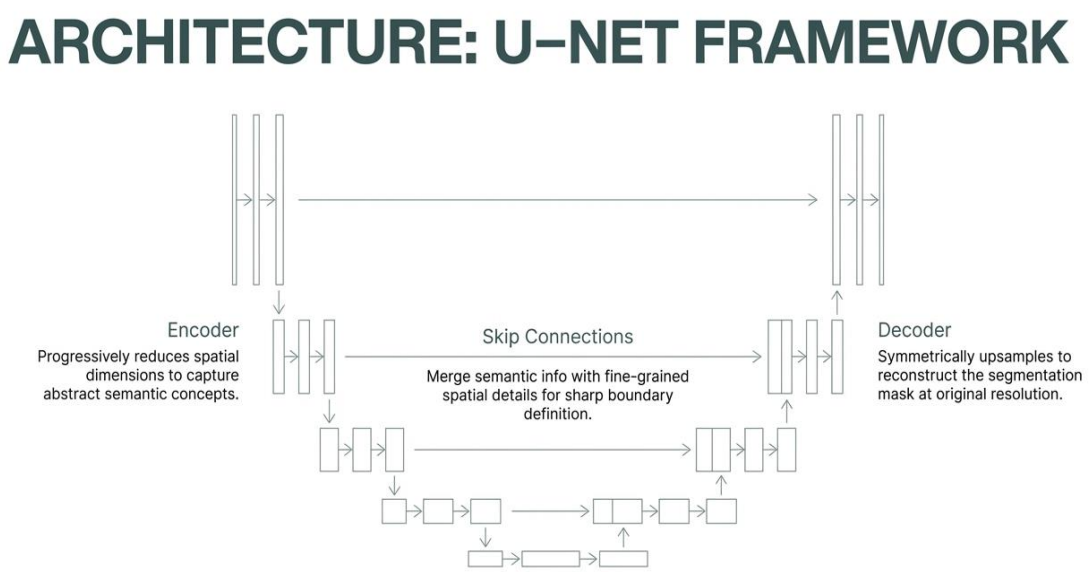
Methodology

U-Net Architecture Utilizes a symmetric encoder-decoder network, widely recognized as highly effective for remote sensing segmentation tasks.

Encoder Path Progressively reduces spatial dimensions to capture abstract, high-level semantic representations of buildings.

Decoder & Skip Connections Upsamples features and merges them with spatial data from the encoder to accurately reconstruct sharp boundaries.

Hybrid Loss Function Combines Binary Cross-Entropy (BCE) for pixel classification with Dice Loss to handle heavy background imbalances.



Data Collection

Inria Aerial Dataset

High-resolution (0.3m/pixel) RGB images from diverse US and Austrian urban areas (residential, industrial).

WHU Building Dataset

Large-scale satellite and aerial imagery featuring precise pixel-level masks and high structural variance.

Massachusetts Dataset

Used primarily as a reserved testing set to evaluate the model's cross-dataset generalization capabilities.

Preprocessing Phase: All images were standardized to 512x512 pixels and normalized to a [0,1] scale to ensure stable and consistent network convergence during training.

Parameters (Overall Works)

Training Optimization

The model leverages the Adam optimizer for adaptive parameter updates.

Initial Learning Rate:
1×10^{-4}

Training Loop: The network processed the combined dataset in batch sizes of 16, iterating over a total of 100 epochs.

Functions & Augmentation

Activations: Non-linear ReLU functions were utilized for all hidden layers to prevent vanishing gradients, while a Sigmoid function was applied to the output layer to generate strict binary probability masks.

Data Augmentation Strategies:

→ Horizontal and vertical flipping

→ Random rotations

→ Brightness and contrast adjustments

Experiment Result

92%

Pixel Accuracy

54%

Intersection over Union

Metric Interpretation

The 92% Pixel Accuracy shows the model successfully classified the vast majority of image pixels. The moderate 54% IoU highlights the inherent difficulty of precise boundary delineation against a massive background class imbalance.

Generalization & Edge Cases

Training on Inria/WHU and testing on Massachusetts proved effective geographic transferability. However, the model faced partial misclassifications in highly dense urban blocks and complex roofs.

Conclusion and Future Works

1. Conclusion

Established a capable, cost-effective segmentation framework utilizing U-Net on accessible RGB imagery.

2. Limitations

Model currently focuses purely on single-image segmentation and needs enhancement for sharper boundary edges.

3. Spatiotemporal

Implement Siamese network architectures to actively compare multi-temporal images for real-time changes.

4. Refinement Edge

Explore boundary refinements (CRFs) and lightweight architectures for edge monitoring.

Reference

Liu, H.; Wang, W.; Tang, J.; Deng, M.; Ding, C. (2025).
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Raj, A.; Mitra, A.; Sinha, M. (2024).
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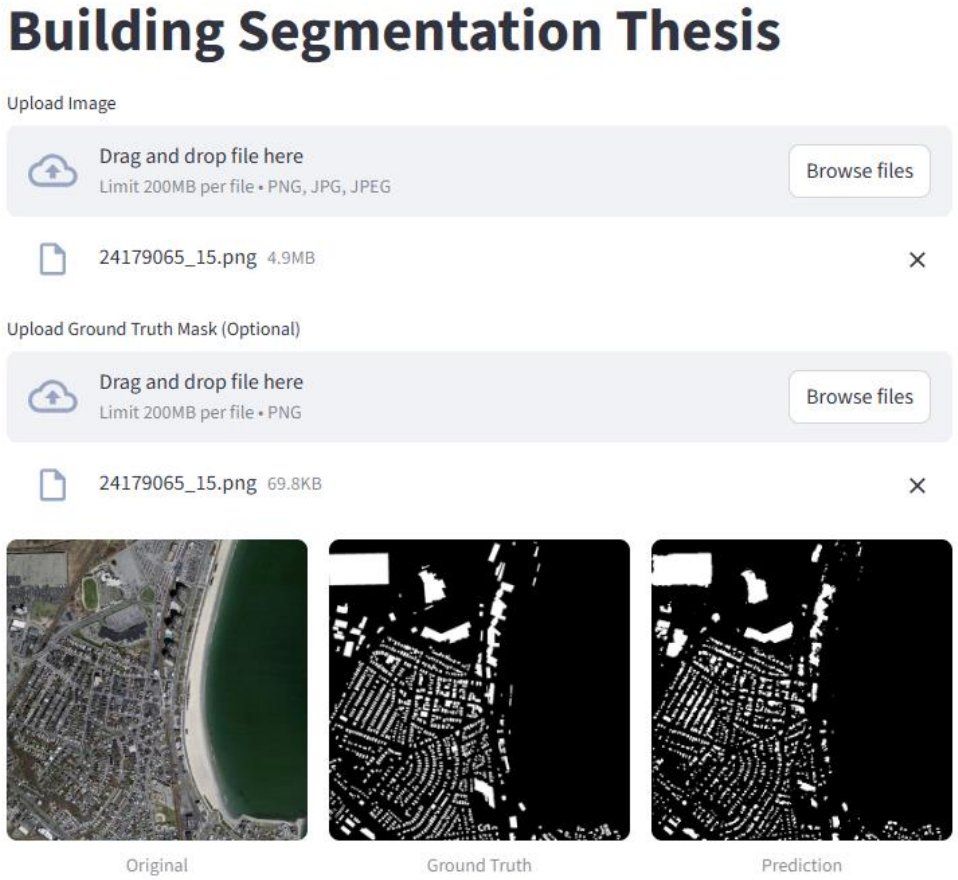
Ji, S., Wei, S., & Lu, M. (2018).
"Fully Convolutional Networks for Multi-Source Building Extraction from An Open Aerial and Satellite Imagery Dataset." IEEE Trans. Geosci. Remote Sens., 57, 574-586.

Website Overview



Thank you

Any question? !?